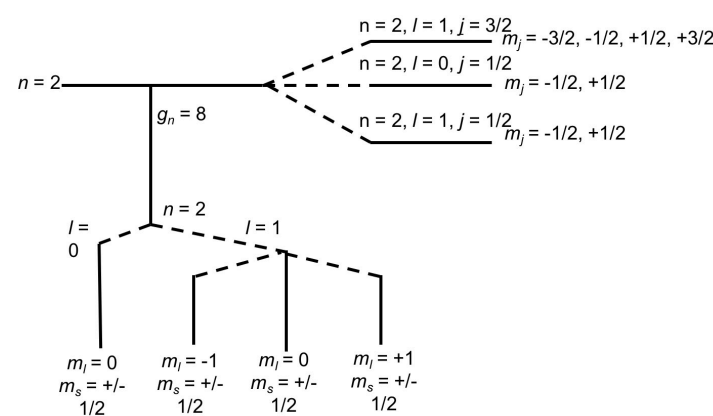


Consider the energy levels of a hydrogenic atom for which $|\lambda_{SO}| \ll \Delta E$ in terms of appropriate quantum number values, list the different eigenstates, first assuming no spin-orbit coupling and then with spin-orbit coupling. Compare the degeneracies for the two cases (including electron spin) and sketch the component levels that result from spin-orbit coupling. Use Fig. 4.33 as a guide.



Your response has been submitted. Engage and discuss with other learners below!

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	Exercise 1	Author: almeida2022 ^[1]	
	File: 1-107-10	File: 1-107-10	Translated to English
Goal	The energy eigenvalues between three sublevels depend on the spin orbit interaction.		
	☐ Clear Plot Author: almeida2022 ^[1] The problem involves computing the energy levels of a hydrogen like atom for $n=3$ and $l=1$ and the interaction		
	Without spin-orbit coupling: The total angular momenta are determined by the quantum numbers l and m . The total degeneracy is given by:		
	$2n^2 = 2 \cdot 3^2 = 18$ With spin-orbit coupling: The total angular momenta is modified as:		
	$2n + l = 6 + 1 = 7$ The quantum number values are:		
	$j = l - s , ..., l + s $ For each j , the possible quantum numbers m_j values are:		
	$m_j = -j, ..., +j$ The degeneracy of each level is given by the energy level and the degeneracy is the sum of all possible quantum numbers m_j for each j and l . To compute the energy levels, we need to solve the energy level equation for spin-orbit coupling. Assuming the levels for $n=3, l=1, s=1/2$ are calculated into sublevels, find m_j and m_l values.		
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